

ICS 406 Module 7 - Project Cost Estimation

What is cost estimation in project management?

Project cost estimation is the process that takes different types of project costs into account and calculates a budget that meets the financial commitment necessary for a successful project.

Project cost estimating involves identifying the tasks, duration, and resources needed to complete the project.

By accurately estimating the cost of a project, a business can better ensure that spending stays within budget once the project begins.

Types of Costs that should be considered

There are five main types of costs that make up the total project cost as shown below:

- **Direct costs:** Direct costs are those that occur in a project and are attached to specific activities of the project. These are generally costs that are easier to accurately estimate. They include raw materials, labour, supplies, etc.
- **Indirect costs:** Indirect costs in a project are those that are in support of the project, such as administrative fees. These can include everything from rent to salaries of the administrative staff to utilities, etc.
- **Fixed costs:** Fixed costs, as the name suggests, are those that do not change throughout the life cycle of a project. Some examples of fixed costs include setup costs, rental costs, insurance premiums, property taxes, etc.
- **Variable costs:** Variable costs are costs that change due to the amount of work that is done in the project and are variable in nature. These costs can include hourly labour wages, materials, fuel costs, and so on.
- **Sunk costs:** In project cost estimating, when an investment has already been incurred and cannot be recovered it is called a sunk cost or retrospective cost. Some examples of sunk costs include marketing, research, installation of new software, etc.

Project costs to consider

A typical project estimate will be made up of the following components:

- **Tasks** – what is going to be done
- **Resources** – who is going to be doing it
- **Duration** – how long it will take
- **Rate** – how much it will cost to deliver the project
- **Tools and software** – what tools and software will be needed

How to estimate project costs in 9 steps

A typical cost estimation process can take the following form:

1. Rework the clients idea
2. Consider conducting a project discovery
3. Break down the project into phases and tasks
4. Calculate the cost for each task
5. Identify if you have enough resources to complete the project
6. Calculate the total cost of the project
7. Incorporate cost implications of risks identified from the risk analysis
8. Add a project buffer to protect profits and allow for risks
9. Account for hidden costs

1. Rework the clients idea

Most clients may not fully understand what they want from a project initially. It may be a basic concept, an end deliverable, or a simple idea. It is your job to rework their idea and turn it into a viable solution you can deliver within their budget.

But what if the client wants to know how much it will cost before committing to working with you? What do you do?

- Get clarity on what you are doing, why, and how. Rework their idea into a viable solution that matches their budget and timeline and delivers on their goals and objectives so that it delivers results.

The basics you need to get information on are based on:

- Client needs – what problems are they facing? How do they expect the project to solve them
- Client expectations – What does success look like to the client? What do the deliverables look like?
- Client budget – What is the client's budget, and what do they want/need to get done for that budget?

Your goal here is to shape their idea into a clear list of deliverables — so that you can articulate what the project is, how it works and how you will deliver it.

2. Consider conducting a project discovery

Gathering all project requirements (or as much as possible) during a project discovery is not just essential, but it is one of the most crucial elements in the project lifecycle.

A project discovery is a process that takes place before kicking off a project. It is focused on collecting detailed project requirements to identify goals, stakeholders, deliverables, costs, and budgets.

When to carry out a project discovery depends on the project's complexity and how much information was collected during the project initiation process. A project discovery could be an initial 30-minute phone call, and that may be sufficient to come up with an estimate of the project's cost, in which case there may be no charge for that.

Otherwise, you could charge for discovery before you commit to the project, then after discovery, refine the project scope and estimate the cost for the project based on the findings from the discovery.

3. Break down the project into phases and tasks

Once you have a solid understanding of what work is required to deliver the project, and the client is happy to move forward with an estimate, a more accurate estimate can be prepared.

The best way to do this is to use a Work Breakdown Structure (WBS) prepared for the project that helps to organize, visualize, and manage projects more efficiently. It serves as a framework for detailed cost estimation and control and can help in building an estimated project timeline.

4. Calculate the cost for each task

Now that you know which tasks you need to complete for the project, you can begin assigning time estimates to each. Team members and clients can work collaboratively to use their knowledge and experience to add more accurate time estimates for each task.

Once a time estimate is available for each task, a cost estimate can be generated based on estimated rates or cost of the resources needed. Past project estimates could provide information that makes budget estimation faster.

5. Identify if you have enough resources to complete the project

A major resource for projects is the human resource. After estimating how many hours you need for each task, consider the team capacity required.

6. Calculate the total cost of the project

Finally, the project's total estimated cost can be calculated using the details in the WBS.

7. Incorporate cost implications of risks identified from the risk analysis

Determine the cost, schedule and technical risks inherent with each item on the WBS.

8. Add a project buffer to protect profits and allow for risks

It is almost impossible to avoid hitches throughout the lifecycle of a project.

The easiest solution is to add a bit of a buffer to the project cost estimates at the task, phase, or project level. Adding this will help account for anything that could go wrong within the project. It will be a much-appreciated cushion for minor issues and sudden changes.

Projects tend to go over budget for reasons such as:

- Additional client requests
- Sudden changes from the client
- Poor project planning
- Goals and objectives that change

Experts recommend adding about 10% – 30% to your project estimate to allow for such.

The better solution to adding buffers to the project cost estimates is to mitigate the impact of risk by estimating for it upfront and adding it as a percent contingency to your estimate to account for change and risk (maybe roundup to no more than 20% of the estimated cost, although many organisations will prefer that this is kept to 10%).

This should be entered as a separate line item in the budget. This helps to highlight to the contractor and client that changes to scope equate to additional costs from the contingency fund. This allows the contractor to say yes to the client a lot more, with the understanding that you will be using the additional budget from the contingency fund for it.

9. Account for hidden costs

Before concluding the cost estimate, hidden costs should be identified and accounted for. Some things to think about are:

Project and account management time

The most significant drain on project costs is the time you spend managing the project and the account. Try adding at most 25% to the project estimate for this. The amount of time you spend on project and account management will vary from project to project. Be realistic and remember that clients want to pay for the work being done so it is better to spread this over the items in the WBS when in doubt.

Scoping time

Remember the time it takes to initiate a project. Think about how and where you will include costs for the effort at the start of the project, pulling together timelines, gathering and compiling project requirements, and putting together the scope of work documentation. Do not bury this in with your estimate — get the client to pay for this as part of an initial discovery phase, or include it in your total estimate as a separate item.

3rd party costs

Finally, do not forget the 3rd party costs. What other things will you need to pay for to deliver the project?

This may include equipment rental, hosting, subscriptions, translations, photography, licensing, and more.

And if working with clients in a different city, remember travel and subsistence and your travel time.

Project estimate vs. budget

What is the difference between an **estimate** and a **project budget**? An estimate is what the cost of a project will be. Once the client approves this, it becomes a project's budget.

Types of estimates

You should follow a simple process of defining the overall project budget and creating different types of estimates before you sign off on a Statement of Work (SoW). A statement of work (SOW) is a document that provides a description of a given project's requirements. It defines the scope of work being provided, project deliverables, timelines, work location, and payment terms and conditions.

Typical project estimates for negotiating and agreeing with clients are, the:

- Ballpark estimate
- Budget estimate
- Statement of Work (SoW) estimate

Ballpark estimate

The client needs to know if the project is possible, but you need more information to give a proper estimate. So you can do this by giving them a ballpark estimate (a term for "rough" estimate) of how much the project will cost (e.g. N80k-N140k).

Unless you have a solid understanding of what needs to be delivered, you should use the top-down or analogous estimate technique to provide a rough ballpark estimate. You do not want to invest too much effort if the client does not have the budget. You can either use your experience here or use data from past similar projects to give a rough estimate.

Budget estimate

Suppose the client is happy with the ballpark estimate. In that case, you may ask for more information about the project to put together a more accurate estimate and a project plan with an estimated timeline. Your revised estimate may change to N90k-N130k.

SoW estimate

Assuming the client is still willing to move ahead, the final step in the estimate refinement is putting together the SoW, which will include the estimate and total project budget and payment terms and conditions. You may adjust your estimate to N100k + a percent contingency fund.

Techniques for estimating project costs

Below are a few cost estimating techniques you can use.

Technique	Description
Analogous estimation	Looks at historical data to create a ballpark estimate
Top-down estimation	Looks at the overall scope and timeframe to create a high-level ballpark estimate
Three-point estimation	Combines the best-case scenario, the worst-case scenario, and the most likely scenario to get a realistic budget estimate
Bottom-up estimation	Breaks down the project into smaller tasks to create a more accurate estimate. Best for SoW estimates

1. Analogous estimation

Analogous estimating involves looking at similar projects you've completed to create your estimate.

For example, a client may want a simple five-page website. You can look at the budget for similar website projects previously completed to create your estimate.

The issue with analogous estimates is that they tend to be slightly inaccurate.

They assume that:

- You have (accurate) historical data to reference
- There are no differences between projects,

But they are quick and easy and can be used with limited information to create a ballpark estimate — especially if you have historical data to hand (more on this later).

2. Top-down estimating

Top-down estimating looks at a high-level project budget and scope to create a ballpark estimate.

The initial project budget should come from the client, and it is your job to take their budget, slice it up, and allocate time/costs to phases or tasks in the project.

You then decide whether or not the client's budget is enough to deliver the project. If not, you can ask the client to adjust the project scope to fit their budget better.

It is a form of analogous estimating, where existing knowledge of similar projects is used in the estimation process to produce a ballpark figure for the total cost.

3. Three-point estimating

Three-point estimating combines the best-case, worst-case, and most likely outcome scenarios to create an accurate, flexible estimate. This technique is particularly useful when the client has asked for a more detailed cost estimate.

How do you use this technique?

The best way is to break the project down into tasks and calculate an hourly estimate for each.

For example, you and your team may estimate the time to create a wireframe as follows:

- Best-case: 4 hours
- Worst-case: 8 hours
- Most likely: 5 hours

Which gives you an estimated time of 5.6 hours.

The main advantage of three-point estimating is that it reduces risk — as you are considering three different scenarios. But the downside is that it takes more time.

4. Bottom-up estimation

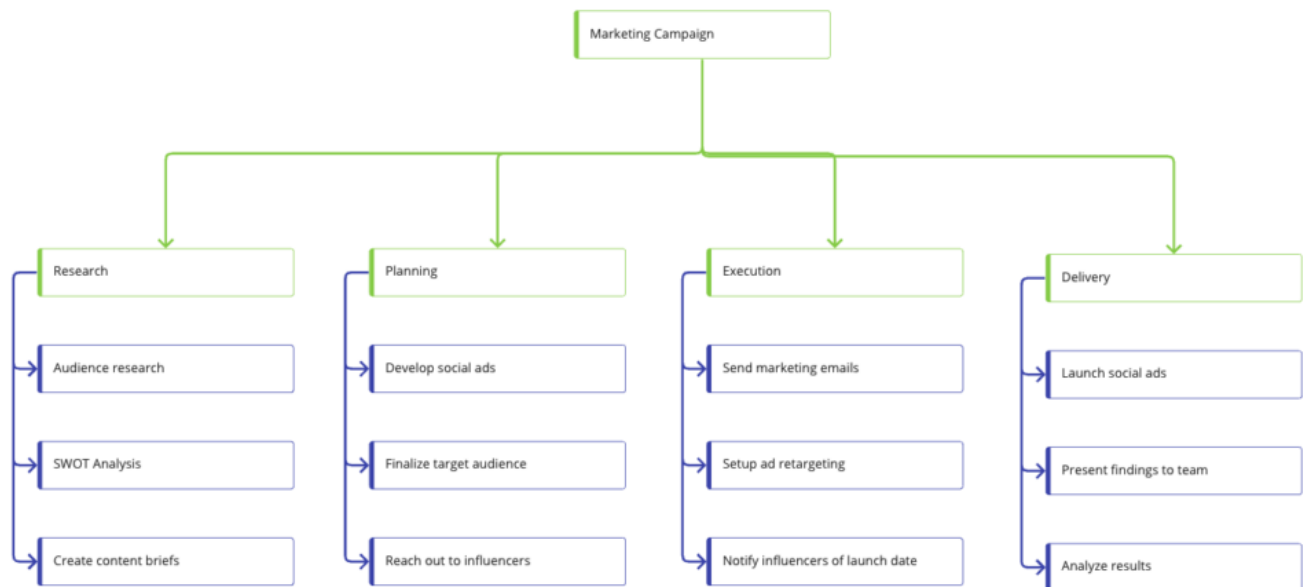
Bottom-up estimating considers the time, cost, and effort for each task within the project using the WBS. It then adds up to create an estimation for the entire project.

The advantage of the bottom-up estimating technique is that it is the most accurate and, by being granular, enables precise tracking of a project's progress against the estimate.

Project cost estimate example

The example is based on a marketing campaign project for a simple explanation of how to estimate project costs.

Using a phase-based approach, the high-level WBS may look like the image below.



Phase-based WBS – Marketing Campaign

Next, you would use the bottom-up estimation technique and calculate how long these tasks will take and determine how much they will cost.

WORK BREAKDOWN STRUCTURE

toggl

Project name	Acme Studios Marketing Campaign Project	Hourly rate	\$150.00
Manager	Bradley	Estimated cost	\$6,600.00

Research				
ID	Tasks	Assigned to	Estimated hours	Estimated cost
1.1	Audience research	Bradley	8	\$1,200.00
1.2	SWOT Analysis	Cecile	4	\$600.00
1.3	Create content briefs	Bradley	7	\$1,050.00
Planning				
ID	Tasks	Assigned to	Estimated hours	Estimated cost
2.1	Develop social ads	Cecile	12	\$1,800.00
2.2	Finalize target audience	Cecile	1	\$150.00
2.3	Reach out to influencers	Cecile	2	\$300.00
Execution				
ID	Tasks	Assigned to	Estimated hours	Estimated cost
3.1	Send marketing emails	Emma	2	\$300.00
3.2	Setup ad retargeting	Emma	2	\$300.00
3.3	Notify influencers of launch date	Team	1	\$150.00
Delivery				
ID	Tasks	Assigned to	Estimated hours	Estimated cost
4.1	Launch social ads	Derek	1	\$150.00
4.2	Present findings to team	Bradley	2	\$300.00
4.3	Analyze results	Team	2	\$300.00

Sub total

\$6,600

Add 10% contingency

660

Grand total

\$7,260